

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 1.0412102665331984, median 0.9076394296732813, std: 0.6775853101247179
Reprojection error (cam1): mean 0.9938837843745534, median 0.843183993368082, std: 0.6793929107413212
Gyroscope error (imu0): mean 1.763013815094, median 1.5372008477792884, std: 1.0234618415487797
Accelerometer error (imu0): mean 0.4645965859245904, median 0.4004885384928988, std: 0.36347265556257635

Residuals

Reprojection error (cam0) [px]: mean 1.0412102665331984, median 0.9076394296732813, std: 0.6775853101247179
Reprojection error (cam1) [px]: mean 0.9938837843745534, median 0.843183993368082, std: 0.6793929107413212
Gyroscope error (imu0) [rad/s]: mean 0.017630138150940004, median 0.015372008477792885, std: 0.010234618415487798
Accelerometer error (imu0) [m/s^2]: mean 0.18583863436983616, median 0.1601954153971595, std: 0.14538906222503054

Transformation (cam0):

T_ci: (imu0 to cam0):

[[-0.00066472 0.02663794 0.99964493 0.01171121]
[-0.00163857 0.99964378 -0.02663899 0.16855436]
[-0.99999844 -0.0016557 -0.00062084 -0.11534674]
[0. 0. 0. 1.]]

T_ic: (cam0 to imu0):

[[-0.00066472 -0.00163857 -0.99999844 -0.11506258]
[0.02663794 0.99964378 -0.0016557 -0.16899726]
[0.99964493 -0.02663899 -0.00062084 -0.00728855]
[0. 0. 0. 1.]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.0002900309601826913

Transformation (cam1):

T_ci: (imu0 to cam1):
[[0.01716141 0.02216055 0.99960712 -0.03696207]
[-0.00526806 0.99974248 -0.02207311 0.1683525]
[-0.99983885 -0.00488719 0.01727373 -0.11662901]
[0. 0. 0. 1.]]

T_ic: (cam1 to imu0):
[[0.01716141 -0.00526806 -0.99983885 -0.115089]
[0.02216055 0.99974248 -0.00488719 -0.16806004]
[0.99960712 -0.02207311 0.01727373 0.04267823]
[0. 0. 0. 1.]]

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
-0.00029670492506792093

Baselines:

Baseline (cam0 to cam1):
[[0.99983109 -0.00450399 -0.01781867 -0.04996747]
[0.0045693 0.99998298 0.00362649 0.0001658]
[0.01780203 -0.00370729 0.99983466 -0.00088495]
[0. 0. 0. 1.]]
baseline norm: 0.049975578152190045 [m]

Gravity vector in target coords: [m/s^2]
[0.26672908 -9.80291914 -0.00740525]

Calibration configuration
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cam0

Camera model: omni
Focal length: [2044.6753301041977, 2065.5927559749084]
Principal point: [616.3310327581959, 404.1255934969876]
Omni xi: 1.5936757817618359
Distortion model: radtan
Distortion coefficients: [2.290964717370836, -1.8671244246514953, -0.010010836103043635, 0.002831387673779597]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.02 [m]
Spacing 0.006 [m]

cam1

Camera model: omni
Focal length: [2037.2418743359228, 2054.1495623160185]
Principal point: [634.1340582275024, 379.92908128448573]
Omni xi: 1.569995877298825
Distortion model: radtan
Distortion coefficients: [2.167702619581351, -0.9560356055374504, -0.01088770643867036, 0.012396986450845738]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.02 [m]
Spacing 0.006 [m]

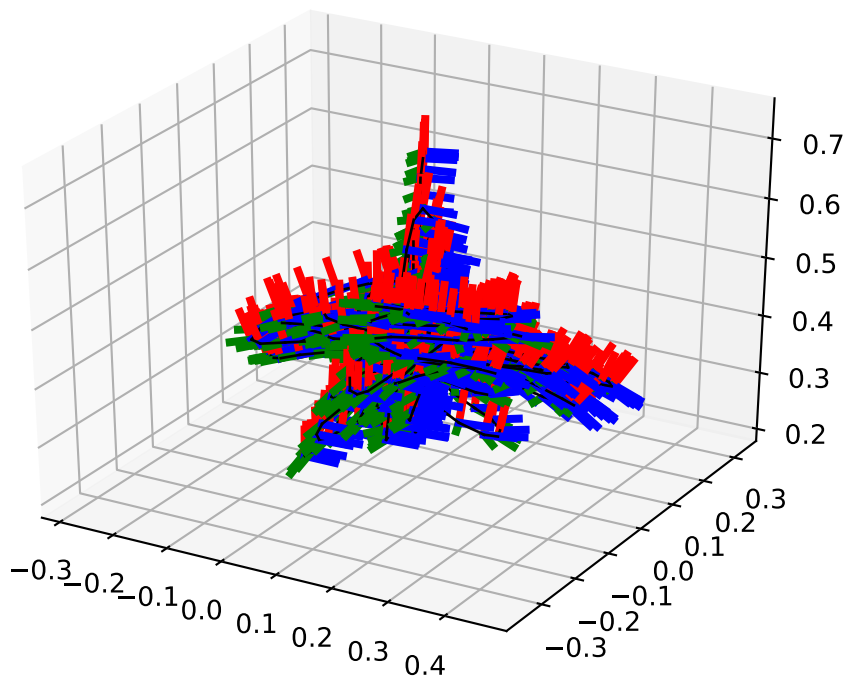
IMU configuration

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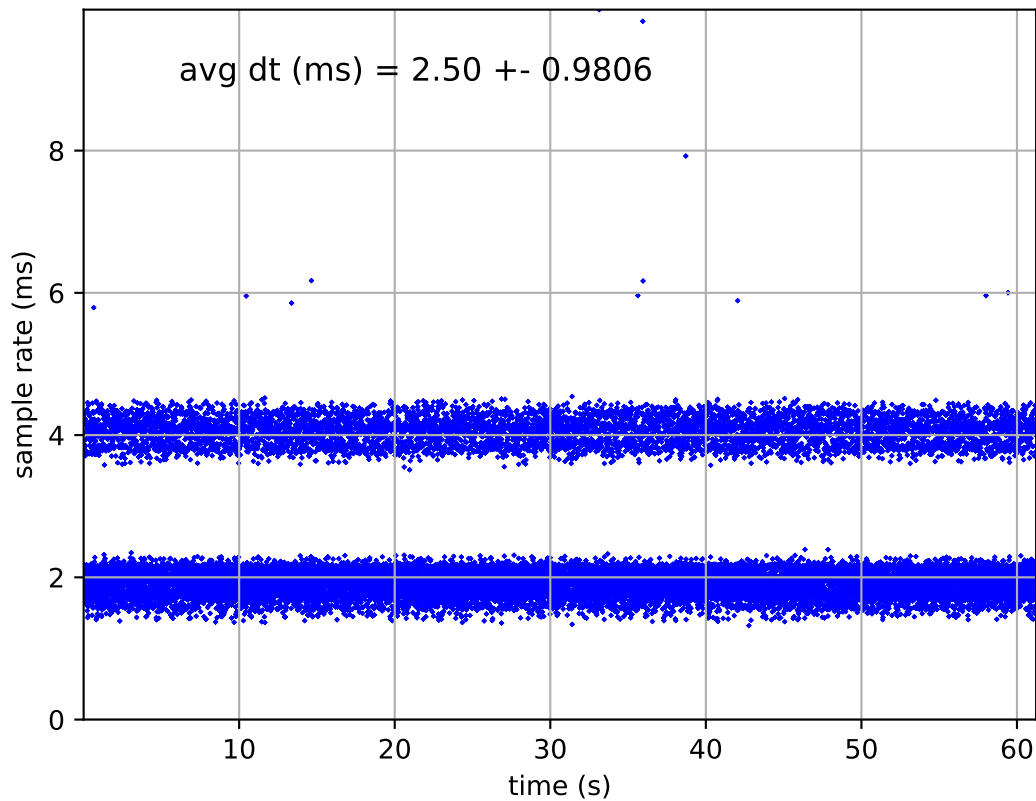
IMU0:

Model: calibrated
Update rate: 400
Accelerometer:
Noise density: 0.02
Noise density (discrete): 0.39999999999999997
Random walk: 0.001
Gyroscope:
Noise density: 0.0005
Noise density (discrete): 0.01
Random walk: 0.0001
T_ib (imu0 to imu0)
[[1. 0. 0. 0.]
[0. 1. 0. 0.]
[0. 0. 1. 0.]
[0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]

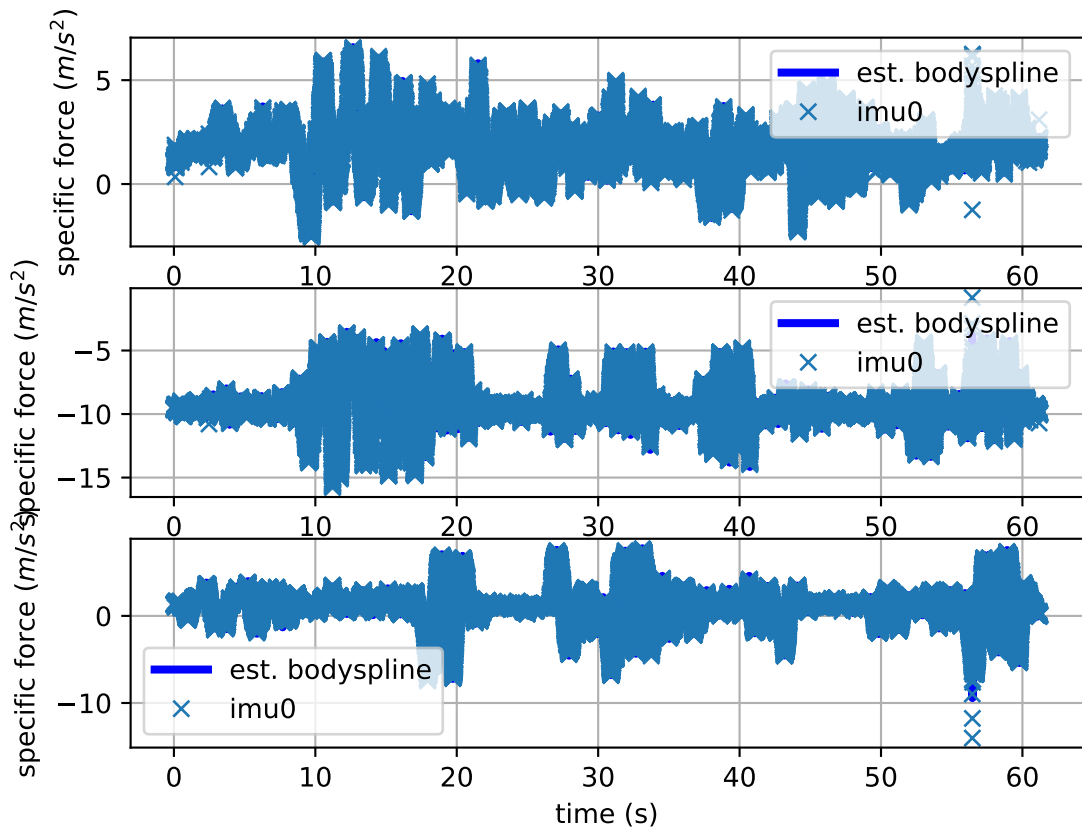
imu0: estimated poses



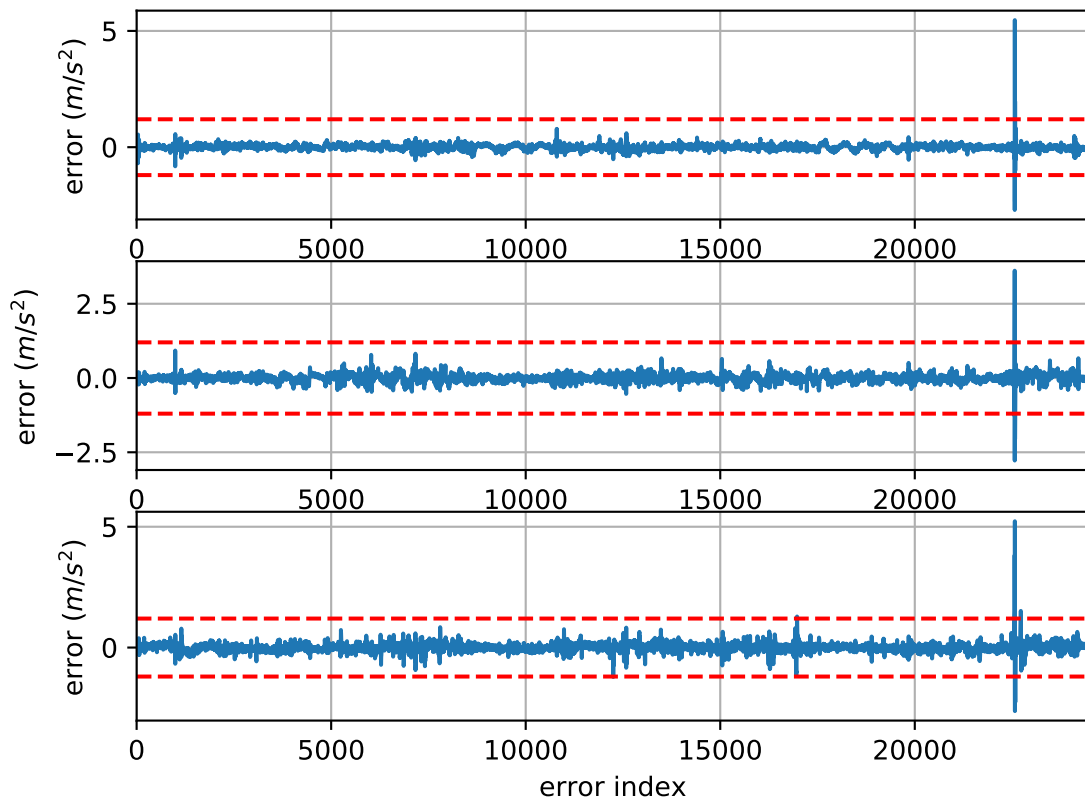
imu0: sample inertial rate



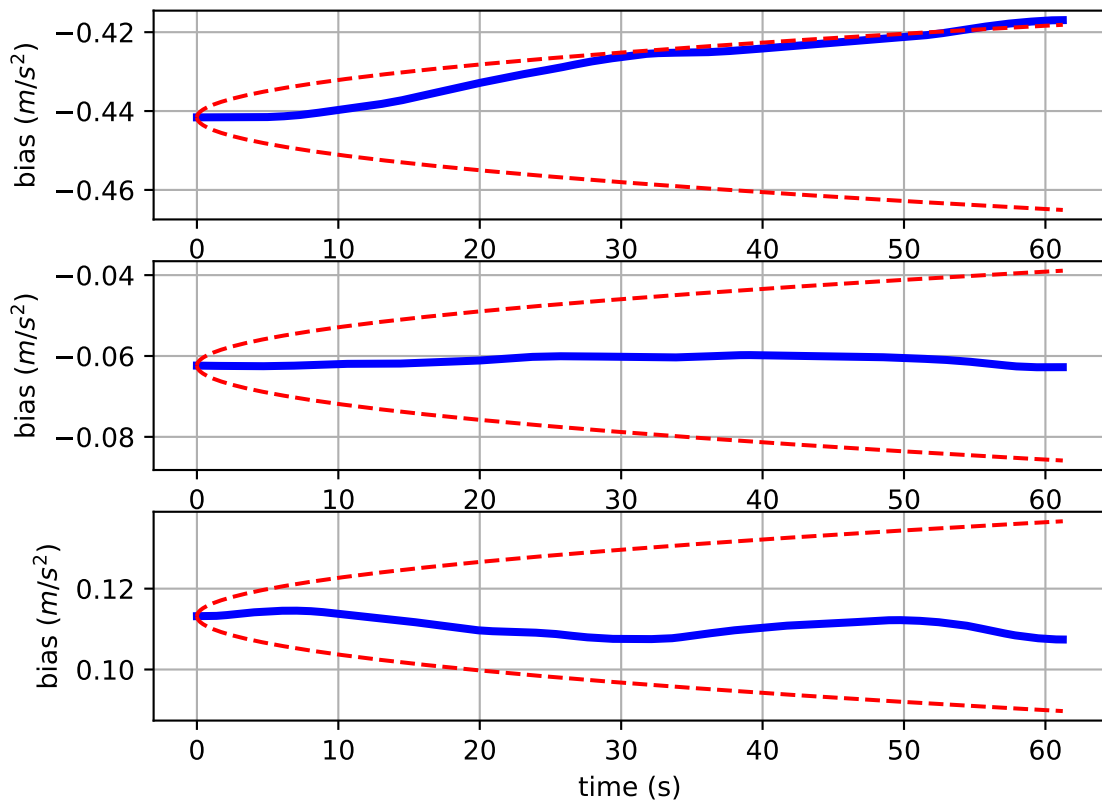
Comparison of predicted and measured specific force (imu0 frame)



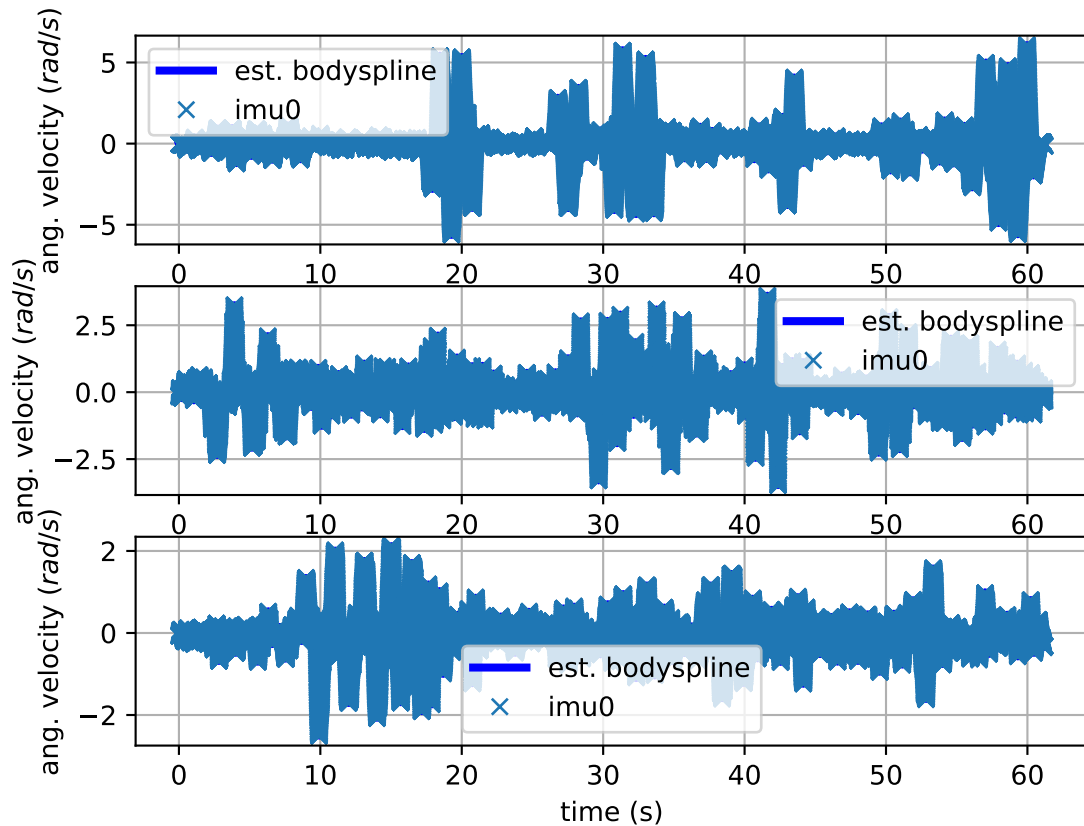
imu0: acceleration error



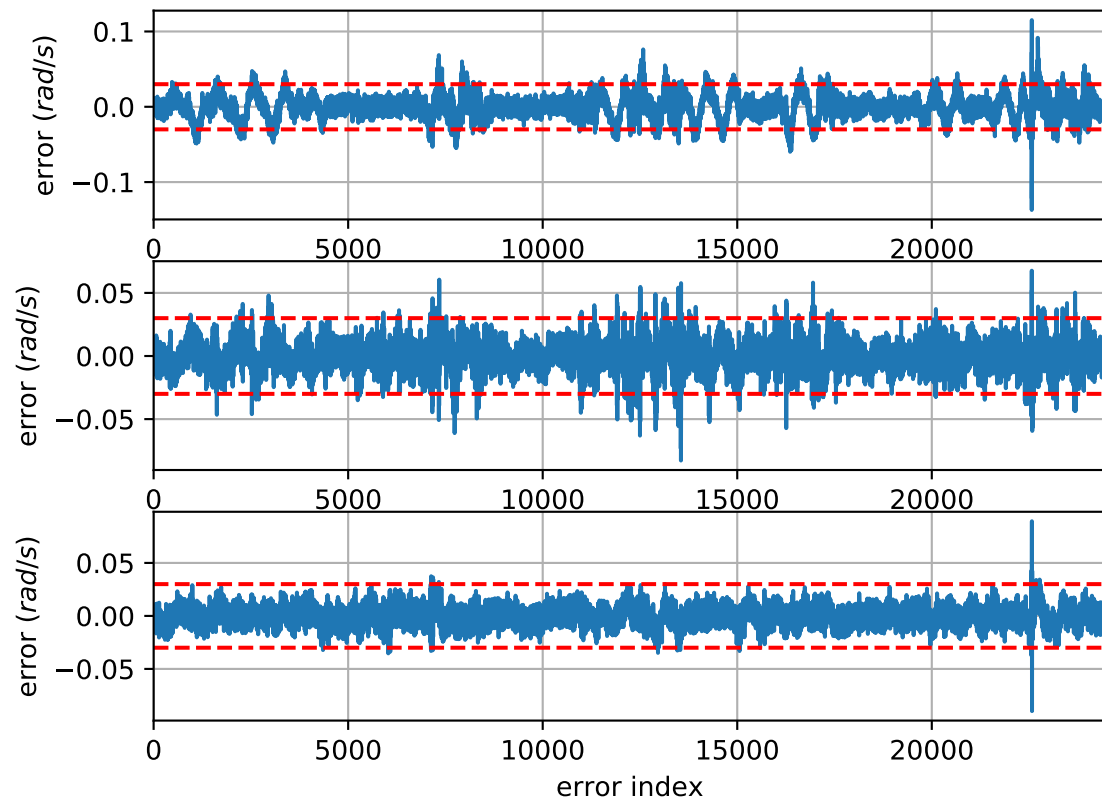
imu0: estimated accelerometer bias (imu frame)



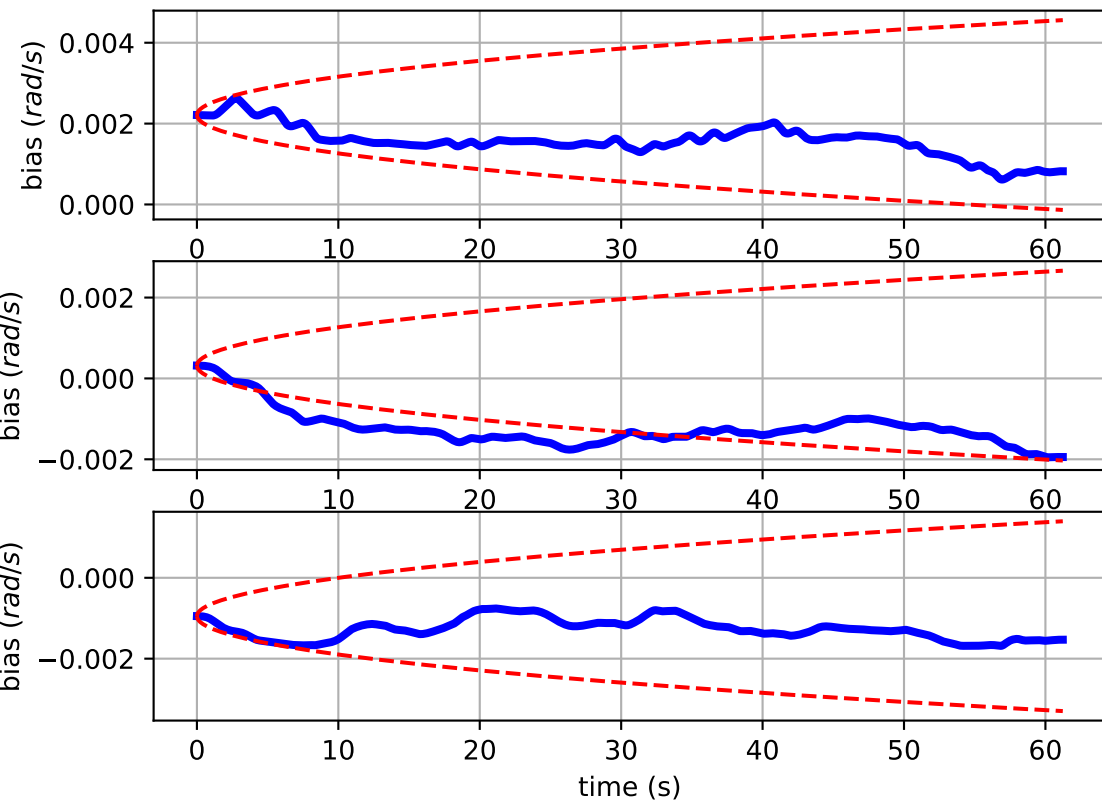
Comparison of predicted and measured angular velocities (body frame)



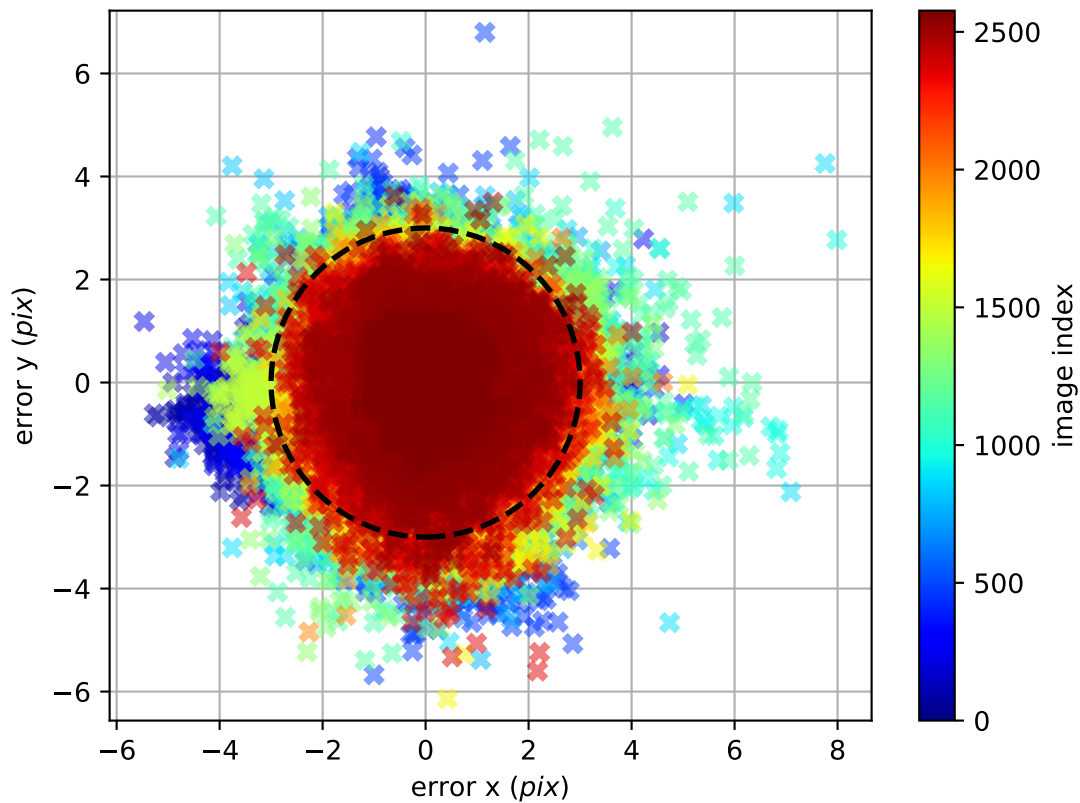
imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors



cam1: reprojection errors

